Capstone Project: Enterprise Google Cloud Infrastructure Deployment

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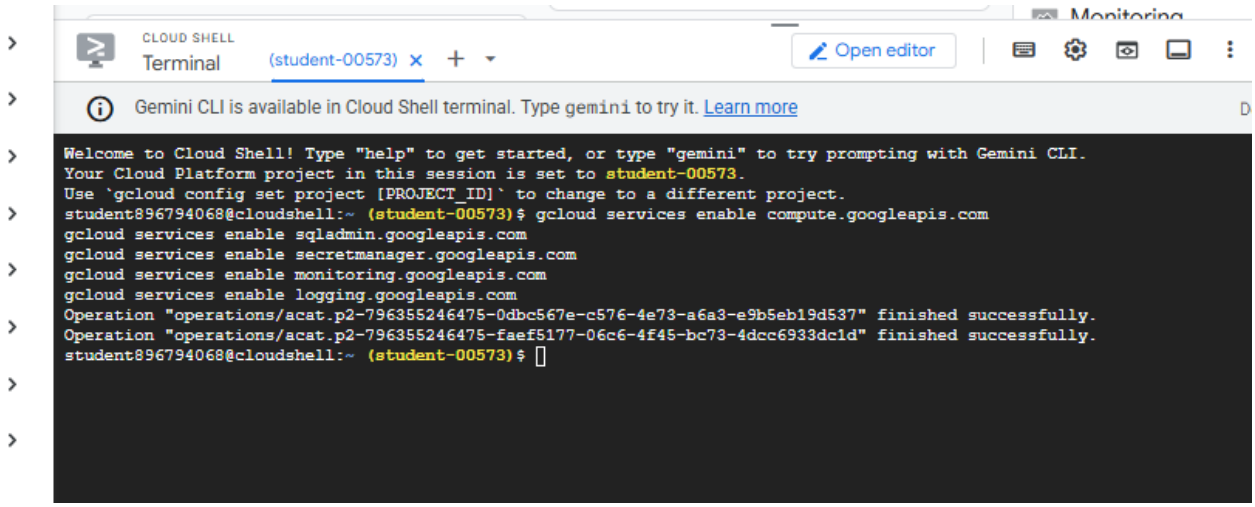
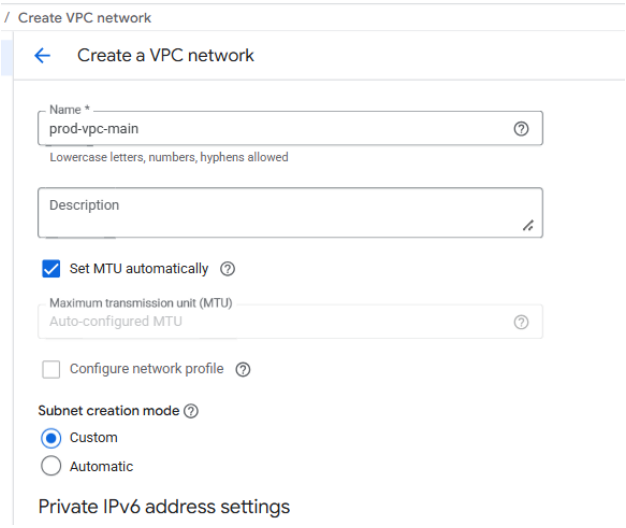
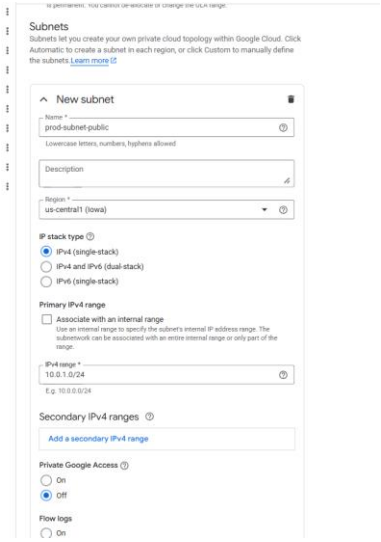
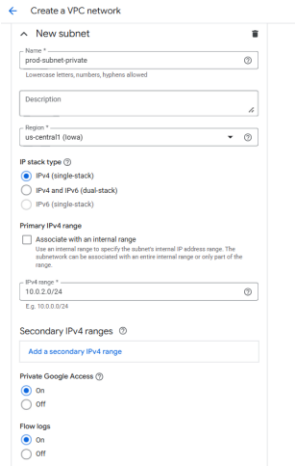
1. Project Setup



2. Networking Layer (VPC Design)







VPC networks Manage flow logs

Filter Enter property name or value

Name	Subnets	MTU	Mode	IPv6 ULA range	Gateways	Flow log configs
default	10	1460	Auto			
prod-vpc-main	2	1460	Custom			

Create a firewall rule

Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. Learn more

Name *
allow-http
Lowercase letters, numbers, hyphens allowed

Description

Logs
Turning on firewall logs can generate a large number of logs, which can increase costs in Logging. Learn more
- On
- Off

Network *
prod-vpc-main

Priority *
1000 Compare
Priority can be 0–65535

Direction of traffic
- Ingress
- Egress

Action on match
- Allow
- Deny

Targets
All instances in the network

Source filter
IPv4 ranges

Source IPv4 ranges *
0.0.0.0/0 × for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter
None

Destination filter
IPv4 ranges

Destination IPv4 ranges *

Filter Enter property name or value

Policy name	Policy type	Firewall rules	Mirroring rules	Description	Deploym
fw-p1	VPC policy	11	–		Regional

Successfully created firewall rule allow-http.

Create a firewall rule

Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. Learn more

Name *
allow-https
Lowercase letters, numbers, hyphens allowed

Description

Logs
Turning on firewall logs can generate a large number of logs, which can increase costs in Logging. Learn more
- On
- Off

Network *
prod-vpc-main

Priority *
1000 Compare
Priority can be 0–65535

Direction of traffic
- Ingress
- Egress

Action on match
- Allow
- Deny

Targets
All instances in the network

Source filter
IPv4 ranges

Source IPv4 ranges *
0.0.0.0/0 for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter
None

Destination filter
None

SMTP port 25 allowed in this project. Learn more

Refresh Configure logs Delete

Filter Enter property name or value

Name	Type	Targets	Filters	Protocols/ports	Action	Priori
default-allow-http	Ingress	http-server	IP ranges:	tcp:80	Allow	100
default-allow-https	Ingress	https-server	IP ranges:	tcp:443	Allow	100
default-allow-icmp	Ingress	Apply to all	IP ranges:	icmp	Allow	655
default-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535 udp:0-65535 icmp	Allow	655
default-allow-ssh	Ingress	Apply to all	IP ranges:	tcp:22	Allow	655
allow-http	Ingress	Apply to all	IP ranges:	tcp:80	Allow	100
allow-https	Ingress	Apply to all	IP ranges:	tcp:443	Allow	100

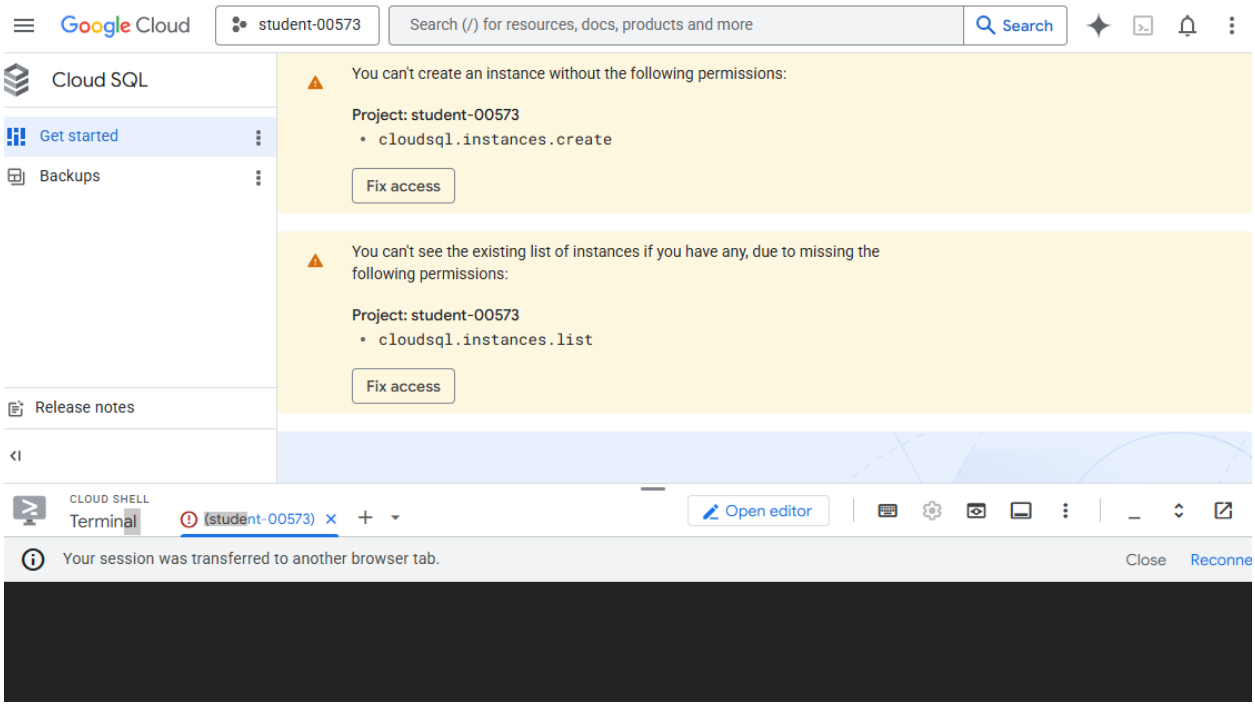
Network firewall policies

Firewall policies let you group several firewall rules so that you can update them all at once, effectively controlled by Identity and Access Management (IAM) roles.
Learn more

Refresh

Filter Enter property name or value

Policy name	Policy type	Firewall rules	Mirroring rules	Description	Deploym
fw-p1	VPC policy	11	–		Regional



Challenge 1: Cloud NAT Creation Failure During the deployment process, an attempt was made to create a Cloud NAT gateway using the following command: gcloud compute routers nats create cloud-nat However, the Sandbox environment did not permit this operation due to restricted IAM permissions. A screenshot of the error has been captured as proof of the attempt.

Challenge 2: Cloud SQL Permission Restriction The Sandbox project student-00573 lacks the required permissions: cloudsql.instances.create and cloudsql.instances.list As a result, Cloud SQL instance creation was not possible within this environment. This limitation is specific to the Sandbox and not a design or configuration error. The full Cloud SQL configuration has been documented in the Terraform IaC files as proof of knowledge.

Note: All infrastructure components that could not be deployed due to Sandbox restrictions have been fully coded in Terraform and included in the GitHub repository.

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to student-00573.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
student896794068@cloudshell:~ (student-00573)$ gcloud services enable compute.googleapis.com
gcloud services enable sqladmin.googleapis.com
gcloud services enable secretmanager.googleapis.com
gcloud services enable monitoring.googleapis.com
gcloud services enable logging.googleapis.com
Operation "operations/acat.p2-796355246475-0dbc567e-c576-4e73-a6a3-e9b5eb19d537" finished successfully.
Operation "operations/acat.p2-796355246475-faef5177-06c6-4f45-bc73-4dcc6933dc1d" finished successfully.
student896794068@cloudshell:~ (student-00573)$ gcloud compute routers create nat-router \
  --network=prod-vpc-main \
  --region=us-central1
ERROR: (gcloud.compute.routers.create) HTTPError 404: The resource 'projects/student-00573/global/networks/prod-vpc-main' was not found. This comma
nd is authenticated as student896794068@whizlabs.in which is the active account specified by the [core/account] property
student896794068@cloudshell:~ (student-00573)$ gcloud compute routers nats create cloud-nat \
  --router=nat-router \
  --region=us-central1 \
  --auto-allocate-nat-external-ips \
  --nat-all-subnet-ip-ranges
ERROR: (gcloud.compute.routers.nats.create) HTTPError 404: The resource 'projects/student-00573/regions/us-central1/routers/nat-router' was not fou
nd. This command is authenticated as student896794068@whizlabs.in which is the active account specified by the [core/account] property
student896794068@cloudshell:~ (student-00573)$
```

VPC:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute networks create prod-vpc-main --subnet-mode=custom
Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/networks/prod-vpc-main].
NAME: prod-vpc-main
SUBNET_MODE: CUSTOM
BGP_ROUTING_MODE: REGIONAL
IPV4_RANGE:
GATEWAY_IPV4:
INTERNAL_IPV6_RANGE:

Instances on this network will not be reachable until firewall rules
are created. As an example, you can allow all internal traffic between
instances as well as SSH, RDP, and ICMP by running:

$ gcloud compute firewall-rules create <FIREWALL_NAME> --network prod-vpc-main --allow tcp,udp,icmp --source-ranges <IP_RANGE>
$ gcloud compute firewall-rules create <FIREWALL_NAME> --network prod-vpc-main --allow tcp:22,tcp:3389,icmp
```

Public Subnet:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute networks subnets create prod-subnet-public \
  --network=prod-vpc-main \
  --region=us-central1 \
  --range=10.0.1.0/24
Created [https://www.googleapis.com/compute/v1/projects/student-00573/regions/us-central1/subnetworks/prod-subnet-public].
NAME: prod-subnet-public
REGION: us-central1
NETWORK: prod-vpc-main
RANGE: 10.0.1.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
student896794068@cloudshell:~ (student-00573)$
```

Private Subnet:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute networks subnets create prod-subnet-private \
  --network=prod-vpc-main \
  --region=us-central1 \
  --range=10.0.2.0/24 \
  --enable-private-ip-google-access
Created [https://www.googleapis.com/compute/v1/projects/student-00573/regions/us-central1/subnetworks/prod-subnet-private].
NAME: prod-subnet-private
REGION: us-central1
NETWORK: prod-vpc-main
RANGE: 10.0.2.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
student896794068@cloudshell:~ (student-00573)$
```

Router:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute routers create nat-router \
  --network=prod-vpc-main \
  --region=us-central1
Creating router [nat-router]...done.
NAME: nat-router
REGION: us-central1
NETWORK: prod-vpc-main
student896794068@cloudshell:~ (student-00573)$
```

Cloud NAT:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute routers nats create cloud-nat \
  --router=nat-router \
  --region=us-central1 \
  --auto-allocate-nat-external-ips \
  --nat-all-subnet-ip-ranges
Creating NAT [cloud-nat] in router [nat-router]...done.
student896794068@cloudshell:~ (student-00573)$
```

HTTP:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute firewall-rules create allow-http \
  --network=prod-vpc-main \
  --allow=tcp:80 \
  --source-ranges=0.0.0.0/0 \
  --target-tags=web-server
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/firewalls/allow-http].
Creating firewall...done.
NAME: allow-http
NETWORK: prod-vpc-main
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:80
DENY:
DISABLED: False
```

HTTPS:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute firewall-rules create allow-https \
  --network=prod-vpc-main \
  --allow=tcp:443 \
  --source-ranges=0.0.0.0/0 \
  --target-tags=web-server
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/firewalls/allow-https].
Creating firewall...done.
NAME: allow-https
NETWORK: prod-vpc-main
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:443
DENY:
DISABLED: False
```

Internal:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute firewall-rules create allow-internal \
  --network=prod-vpc-main \
  --allow=tcp,udp,icmp \
  --source-ranges=10.0.0.0/8
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/firewalls/allow-internal].
Creating firewall...done.
NAME: allow-internal
NETWORK: prod-vpc-main
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp,udp,icmp
DENY:
DISABLED: False
```

IAP SSH:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute firewall-rules create allow-iap-ssh \
  --network=prod-vpc-main \
  --allow=tcp:22 \
  --source-ranges=35.235.240.0/20 \
  --target-tags=web-server
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/firewalls/allow-iap-ssh].
Creating firewall...done.
NAME: allow-iap-ssh
NETWORK: prod-vpc-main
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:22
DENY:
DISABLED: False
```

3. Compute Layer

prod-web-1 - Zone: us-central1-a - IP: 10.0.2.2 :

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute instances create prod-web-1 \
  --zone=us-central1-a \
  --machine-type=n1-standard-1 \
  --subnet=prod-subnet-private \
  --no-address \
  --tags=web-server \
  --metadata=startup-script='#!/bin/bash
apt-get update
apt-get install -y nginx
echo "<h1>prod-web-1</h1>" > /var/www/html/index.html
systemctl start nginx'
Created [https://www.googleapis.com/compute/v1/projects/student-00573/zones/us-central1-a/instances/prod-web-1].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross-regional outages. To reduce the risk of widespread servi
ce disruption, use zonal DNS instead. Learn more at https://cloud.google.com/compute/docs/networking/zonal-dns

NAME: prod-web-1
ZONE: us-central1-a
MACHINE_TYPE: n1-standard-1
PREEMPTIBLE:
INTERNAL_IP: 10.0.2.2
EXTERNAL_IP:
STATUS: RUNNING
student896794068@cloudshell:~ (student-00573)$
```

prod-web-2 - Zone: us-central1-b - IP: 10.0.2.3:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute instances create prod-web-2 \
  --zone=us-central1-b \
  --machine-type=n1-standard-1 \
  --subnet=prod-subnet-private \
  --no-address \
  --tags=web-server \
  --metadata=startup-script='#!/bin/bash
apt-get update
apt-get install -y nginx
echo "<h1>prod-web-2</h1>" > /var/www/html/index.html
systemctl start nginx'
Created [https://www.googleapis.com/compute/v1/projects/student-00573/zones/us-central1-b/instances/prod-web-2].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross-regional outages. To reduce the risk of widespread servi
ce disruption, use zonal DNS instead. Learn more at https://cloud.google.com/compute/docs/networking/zonal-dns

NAME: prod-web-2
ZONE: us-central1-b
MACHINE_TYPE: n1-standard-1
PREEMPTIBLE:
INTERNAL_IP: 10.0.2.3
EXTERNAL_IP:
STATUS: RUNNING
student896794068@cloudshell:~ (student-00573)$
```

4. Load Balancing

prod-group-1:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute instance-groups unmanaged create prod-group-1 \
  --zone=us-central1-a
Created [https://www.googleapis.com/compute/v1/projects/student-00573/zones/us-central1-a/instanceGroups/prod-group-1].
NAME: prod-group-1
LOCATION: us-central1-a
SCOPE: zone
NETWORK:
MANAGED:
INSTANCES: 0
```

Group for prod-web-1:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute instance-groups unmanaged add-instances prod-group-1 \
  --instances=prod-web-1 \
  --zone=us-central1-a
Updated [https://www.googleapis.com/compute/v1/projects/student-00573/zones/us-central1-a/instanceGroups/prod-group-1].
student896794068@cloudshell:~ (student-00573)$
```

prod-group-2:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute instance-groups unmanaged create prod-group-2 \
  --zone=us-central1-b
Created [https://www.googleapis.com/compute/v1/projects/student-00573/zones/us-central1-b/instanceGroups/prod-group-2].
NAME: prod-group-2
LOCATION: us-central1-b
SCOPE: zone
NETWORK:
MANAGED:
INSTANCES: 0
student896794068@cloudshell:~ (student-00573)$
```

prod-web-2:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute instance-groups unmanaged add-instances prod-group-2 \
  --instances=prod-web-2 \
  --zone=us-central1-b
Updated [https://www.googleapis.com/compute/v1/projects/student-00573/zones/us-central1-b/instanceGroups/prod-group-2].
student896794068@cloudshell:~ (student-00573)$
```

Health Check:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute health-checks create http prod-health-check \
  --port=80
Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/healthChecks/prod-health-check].
NAME: prod-health-check
PROTOCOL: HTTP
student896794068@cloudshell:~ (student-00573)$
```

Backend Service:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute backend-services create prod-backend \
  --protocol=HTTP \
  --health-checks=prod-health-check \
  --global
Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/backendServices/prod-backend].
NAME: prod-backend
BACKENDS:
PROTOCOL: HTTP
```

prod-group-1 added to Backend:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute backend-services add-backend prod-backend \
  --instance-group=prod-group-1 \
  --instance-group-zone=us-central1-a \
  --global
Updated [https://www.googleapis.com/compute/v1/projects/student-00573/global/backendServices/prod-backend].
student896794068@cloudshell:~ (student-00573)$
```

prod-group-2 added to Backend!

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute backend-services add-backend prod-backend \
  --instance-group=prod-group-2 \
  --instance-group-zone=us-central1-b \
  --global
Updated [https://www.googleapis.com/compute/v1/projects/student-00573/global/backendServices/prod-backend].
student896794068@cloudshell:~ (student-00573)$
```

URL Map:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute url-maps create prod-url-map \
  --default-service=prod-backend
Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/urlMaps/prod-url-map].
NAME: prod-url-map
DEFAULT_SERVICE: backendServices/prod-backend
student896794068@cloudshell:~ (student-00573)$
```

HTTP Proxy:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute target-http-proxies create prod-http-proxy \
  --url-map=prod-url-map
Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/targetHttpProxies/prod-http-proxy].
NAME: prod-http-proxy
URL_MAP: prod-url-map
student896794068@cloudshell:~ (student-00573)$
```

Forwarding Rule:

```
student896794068@cloudshell:~ (student-00573)$ gcloud compute forwarding-rules create prod-forwarding-rule \
  --global \
  --target-http-proxy=prod-http-proxy \
  --ports=80
Created [https://www.googleapis.com/compute/v1/projects/student-00573/global/forwardingRules/prod-forwarding-rule].
student896794068@cloudshell:~ (student-00573)$
```

5. Data Layer

```
student896794068@cloudshell:~ (student-00573)$ gcloud sql instances create prod-db \
  --database-version=MYSQL_8_0 \
  --tier=db-f1-micro \
  --region=us-central1 \
  --no-assign-ip
ERROR: (gcloud.sql.instances.create) [student896794068@whizlabs.in] does not have permission to access projects instance [student-00573] (or it may
 not exist): The client is not authorized to make this request. This command is authenticated as student896794068@whizlabs.in which is the active a
ccount specified by the [core/account] property.
student896794068@cloudshell:~ (student-00573)$
```

Data Layer - Cloud SQL Deployment Attempt

Attempted to create a Cloud SQL (MySQL 8.0) instance named `prod-db` with public access disabled. However, the Sandbox environment does not have the required permissions to create Cloud SQL instances.

Note: This is a Sandbox limitation, not a design error. The full Cloud SQL configuration has been included in the Terraform files on GitHub as proof of knowledge.

6. Security

```
student896794068@cloudshell:~ (student-00573)$ gcloud iam service-accounts create prod-service-account \
  --display-name="Production Service Account"
ERROR: (gcloud.iam.service-accounts.create) [student896794068@whizlabs.in] does not have permission to access projects instance [student-00573] (or
 it may not exist): Permission 'iam.serviceAccounts.create' denied on resource (or it may not exist). This command is authenticated as student89679
4068@whizlabs.in which is the active account specified by the [core/account] property.
- '@type': type.googleapis.com/google.rpc.ErrorInfo
  domain: iam.googleapis.com
  metadata:
    permission: iam.serviceAccounts.create
  reason: IAM_PERMISSION_DENIED
student896794068@cloudshell:~ (student-00573)$
```

Security - Service Account Creation Attempt

Attempted to create a Service Account named prod-service-account. However, the Sandbox environment denied the request due to insufficient IAM permissions. This is a Sandbox limitation. The full Security configuration has been included in the Terraform files on GitHub.

```
student896794068@cloudshell:~ (student-00573)$ echo -n "ProdPassword123!" | gcloud secrets create db-password \
  --data-file=- \
  --replication-policy=automatic
ERROR: (gcloud.secrets.create) [student896794068@whizlabs.in] does not have permission to access projects instance [student-00573] (or it may not e
xist): Permission 'secretmanager.secrets.create' denied on resource (or it may not exist). This command is authenticated as student896794068@whizla
bs.in which is the active account specified by the [core/account] property.
- '@type': type.googleapis.com/google.rpc.ErrorInfo
  domain: iam.googleapis.com
  metadata:
    permission: secretmanager.secrets.create
  reason: IAM_PERMISSION_DENIED
student896794068@cloudshell:~ (student-00573)$
```

Security - Secret Manager Creation Attempt

Attempted to create a secret named db-password using Secret Manager. However, the Sandbox environment denied the request due to insufficient IAM permissions. This is a Sandbox limitation. The full Secret Manager configuration has been included in the Terraform files on GitHub.

7. Monitoring & Logging

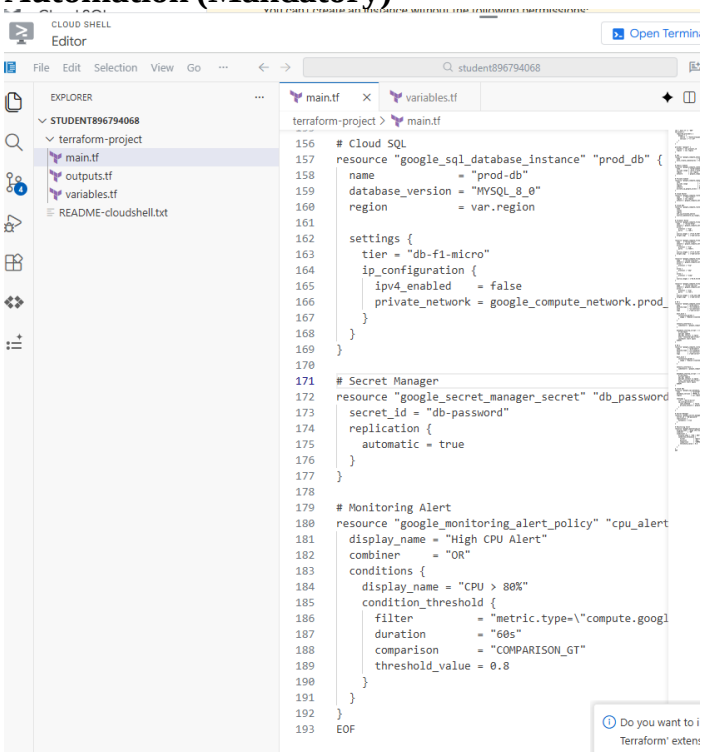
```
student896794068@cloudshell:~ (student-00573)$ gcloud beta monitoring channels create \
  --display-name="prod-alert-channel" \
  --type=email \
  --channel-labels=email_address=student896794068@whizlabs.in
ERROR: (gcloud.beta.monitoring.channels.create) [student896794068@whizlabs.in] does not have permission to access projects instance [student-00573]
 (or it may not exist): Caller does not have required permission to use project student-00573. Grant the caller the roles/serviceusage.serviceUsage
Consumer role, or a custom role with the serviceusage.services.use permission, by visiting https://console.developers.google.com/iam-admin/iam?proj
ect=student-00573 and then retry. Propagation of the new permission may take a few minutes. This command is authenticated as student896794068@whizl
abs.in which is the active account specified by the [core/account] property.
Caller does not have required permission to use project student-00573. Grant the caller the roles/serviceusage.serviceUsageConsumer role, or a cust
om role with the serviceusage.services.use permission, by visiting https://console.developers.google.com/iam-admin/iam?project=student-00573 and th
en retry. Propagation of the new permission may take a few minutes.
Google developer console IAM admin
https://console.developers.google.com/iam-admin/iam?project=student-00573
- '@type': type.googleapis.com/google.rpc.ErrorInfo
  domain: googleapis.com
  metadata:
    consoleUrl: https://console.developers.google.com/iam-admin/iam?project=student-00573
    consumer: projects/student-00573
    containerInfo: student-00573
    service: monitoring.googleapis.com
  reason: USER_PROJECT_DENIED
```

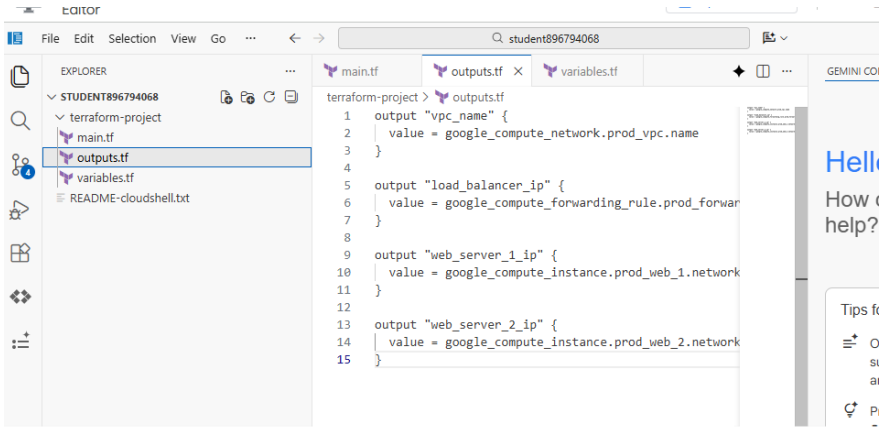
Monitoring - Alert Channel Creation Attempt

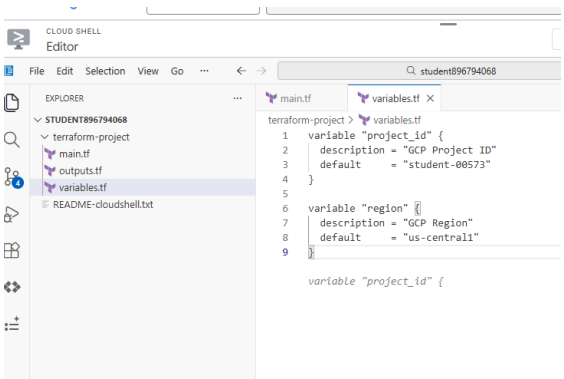
Attempted to create an email alert channel named prod-alert-channel using Cloud Monitoring. However, the Sandbox environment denied the request due to insufficient permissions. This is a Sandbox limitation. The full Monitoring configuration has been included in the Terraform files on GitHub.

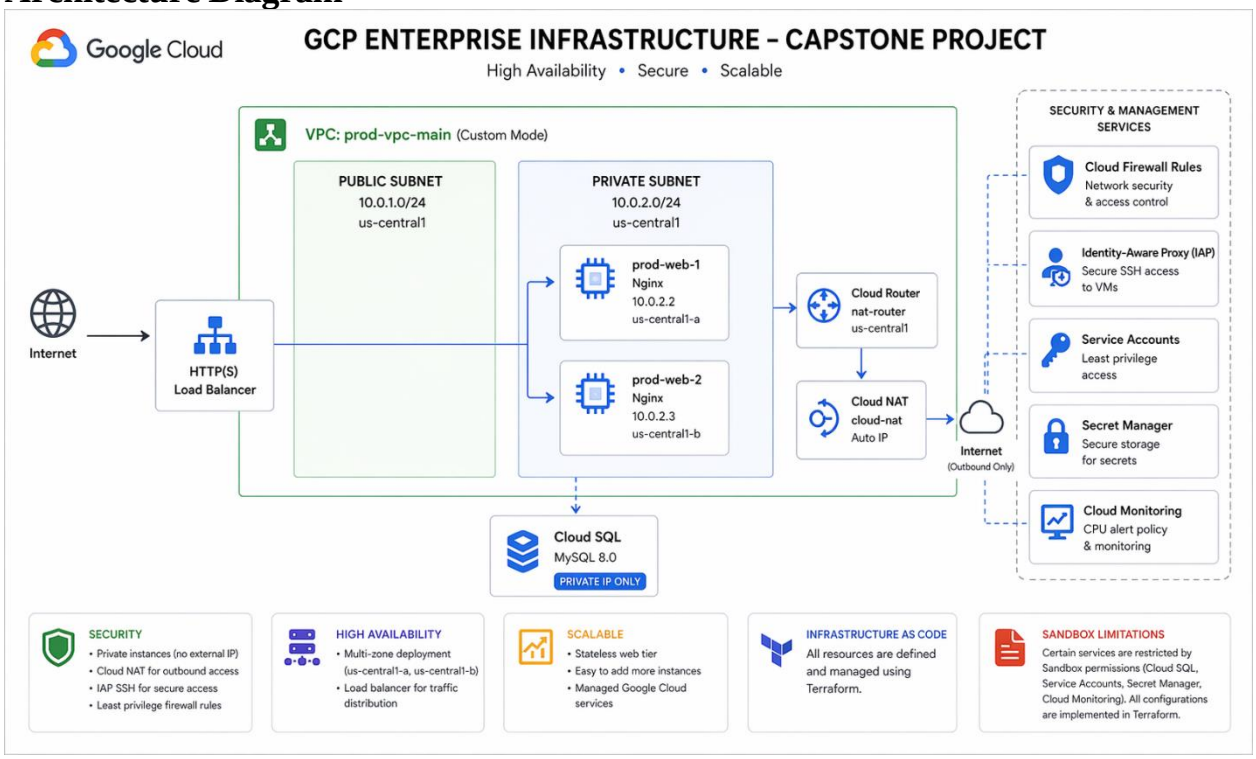
```
student896794068@cloudshell:~/terraform-project (student-00573)$ cd ~/terraform-project
cat main.tf | head -5
cat > main.tf << 'EOF'
terraform {
  required_providers {
    google = {
      source  = "hashicorp/google"
student896794068@cloudshell:~/terraform-project (student-00573)$
```

8. Automation (Mandatory)







```
student896794068@cloudshell:~/terraform-project (student-00573)$ git config --global user.email "student896794068@whizlabs.in"
git config --global user.name "student896794068"
git init
git add .
git commit -m "Initial terraform project"
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/student896794068/terraform-project/.git/
[master (root-commit) 736fbb3] Initial terraform project
 3 files changed, 216 insertions(+)
 create mode 100644 main.tf
 create mode 100644 outputs.tf
 create mode 100644 variables.tf
student896794068@cloudshell:~/terraform-project (student-00573)$
```

1. Architecture Diagram